



Building Certification | Building Code Consultancy

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Job Ref. KDS-Q258339

Rev 1 – 3rd of September 2025

Rev 2 - 12th of September 2025

Rev 3 – 21st of January 2026

Rev 4 – 5th of February 2026

Brisbane Airport Corporation (BAC)
11 The Circuit,
Brisbane Airport,
QLD 4008

Attention: Emil Bighi,

Dear Emil,

RE: BUILDING CODE ASSESSMENT REPORT for BAC BNE Play (AREA 51) and Game Over, amenities and carpark – 11 The Circuit, Brisbane Airport QLD 4008 – Lot 10 SP238997

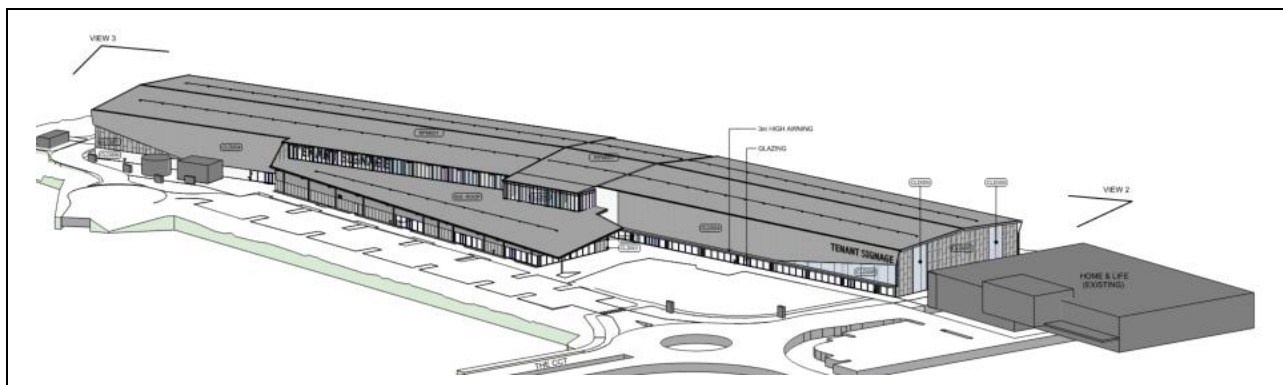
Following a review of the latest design documentation, Knisco Development Solutions (Knisco) provide an assessment report outlining design matters to be considered during the design development for the proposed BAC BNE Play (AREA 51 & Game Over), 11 The Circuit, Brisbane Airport QLD 4008 (Lot 10 SP238997). The recommendations within this report should be addressed prior to submitting a Building Application to Airport Building Control (ABC).

If you have any queries, please do not hesitate to contact me on 07 3852 2080.

Regards,

Tracey McLean
Building Surveyor – Knisco Development Solutions Pty Ltd

BUILDING CODE ASSESSMENT REPORT



Job Reference:	KDS-Q258339
Project:	BAC BNE Play
Project Address:	11 The Circuit, Brisbane Airport QLD 4009, Lot 10 SP238997
Applicable BCA Edition:	BCA 2022 Amendment 2
Revision 1	03 September 2025
Revision 2	12 September 2025
Revision 3	21 st of January 2026
Revision 4	5 th of February 2026
Prepared by:	Tracey McLean
Reviewed by	Franco Valasco

1. INTRODUCTION

Knisco Development Solutions (Knisco) have been engaged by Brisbane Airport Corporation Pty Ltd to review the preliminary design documentation for the proposed BAC BNE Play located at 11 The Circuit, Brisbane Airport QLD 4008.

The proposed work is understood to be a new warehouse buildings (AREA 51 & Game Over), amenities and carpark as per the latest documentation received on the 5th of February 2026 prepared by Archipelago Architects. It is further understood the proposed work being a new warehouse (AREA 51 & Game Over), will include Performance Solutions for departures from the BCA Deemed-to-Satisfy (DTS) Provisions.

Any reference to the BCA is a reference to the National Construction Code Building Code of Australia Volume One 2022 Amendment 2.

2. BUILDING DESCRIPTION

The following building description is understood:

Proposed/Intended use	Play centre & Game Over
Building Classification	Class 9b (playcentre AREA 51) Class 9b (Game Over) Class 6 (café)
Type of Construction	Type C (Large Isolated Building)
Rise in Storeys	1
Number of storeys	1
Effective Height	0m
Floor Area of Building	Approx. 10,715m ²
Size of Largest Fire Compartment	The building will be fire separated into 2 separate fire compartments with the largest being the Play Centre AREA 51 which will have the following: Floor area - Approx. 7,115m ² Volume - Approx. 71,810m ³
Energy Efficiency Zone	Zone 2

3. PERFORMANCE SOLUTIONS

Following this detailed assessment, Knisco have identified the following areas that may be capable of being assessed against the Performance Requirements of the BCA as a Performance Solution (in lieu of in accordance with the Deemed-to-Satisfy Provisions).

NOTE: Further discussion with all relevant stakeholders will be required prior to any Performance Solutions being accepted. It is recommended that these discussions commence as soon as possible to avoid any delays. We recommend consultation with Airport Building Control prior to proceeding with Performance Solutions.

No.	Description of Performance Solution	DTS Clause/s	Performance Requirement/s
1.	Permit the furthest boundary of the perimeter access road to be up to 52m away from the building (a large isolated building) in lieu of max 18m from the building	C3D5 & C3D5 (2)	C1P9
2.	Permit exposure to openings in the external walls of separate fire compartments to be rationalized as a fire engineered performance solution.	C4D4	C1P2
3.	In order to 'future proof' any tenancy fitout layouts the fire engineer has justified the following travel distances: 30m in lieu of max 20m to a point of choice between alternative exits 60m in lieu of max 40m to an exit	D2D5	D1P4 & E2P2
4.	The base building (open plan design) has extended distance between exits of 67m between alternative exits in lieu of max 60m as an open plan layout and an additional exit door	D2D6	D1P4 & E2P2

	required or fire engineered performance solution required. In order to 'future proof' any tenancy fitout layouts the fire engineer has justified the following distance between alternative exits: 90m between alternative exits in lieu of max 60m		
5.	Permit variations to AS 2419.1 (2021) as follows: - If internal hydrants are installed/required more than 4m from an exit they must be documented as a fire engineered performance solution – confirmation required of location of fire hydrants. - The location of the booster assembly is not in the façade of the building containing the principal pedestrian entrance.	E1D2 & AS 2419.1 (2021)	E1P3
6.	Permit omission of sprinklers protection below play equipment exceeding 800 mm width within the Area 51, in lieu of complying with AS 2118.1, which requires additional sprinklers under obstruction of this width.	E1D4	E1P4
7.	Permit directional exits signage to be installed at a height exceeding 2.7m high.	E4D5	E4P2
8.	Compliance with F3P1 for the weatherproofing of an external wall to be documented as a performance solution.	F3D5	F3P1
9.	Permit compliance with JP1 to be verified using JV3 - Verification using a reference building – confirm if applicable.	-	J1P1
10.	Permit sanitary facilities basins to be unisex and located in a communal area in lieu of being dedicated male basins and female basins.	F4D4	F4P1

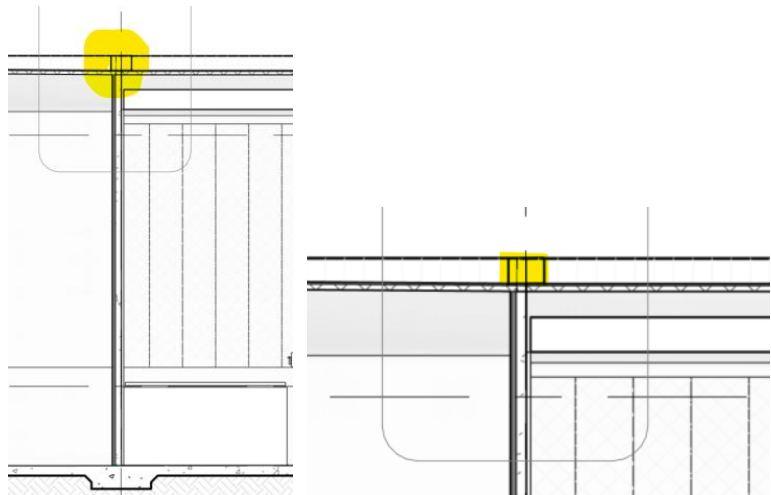
4. BUILDING CODE ASSESSMENT REPORT (Building Code of Australia 2022 Amendment 2)

The following comments are offered in regard to the proposed new warehouse building. We recommend any non-compliance matters listed below be addressed during design development prior to submitting a Building Application to Airport Building Control (ABC).

Item	Reference	Issue	Recommendation
4.1. Section B – Structure			
<u>Part B1 – Structural Provisions</u>			
1.	BCA Part B1D2 & B1D3 & D2D4	All structural aspects of the building (and ancillary structures) must be compliant with Part B1 of the BCA & AS 1170. Knisco understand the importance level of the building will be importance level 3 given more than 300 persons can congregate in one area (The importance level must be confirmed by the structural engineer).	Design certificate to ABC's template required for submission to ABC.
2.	BCA Part B1D3	Non-structural parts of the building & services must be designed & constructed to comply with AS1170.4 (2007) for earthquake loads. For the purpose of this project, these parts & services to include: - Walls, partitions & ceilings - Mechanical ventilation equipment - Fire services infrastructure.	Design certificate to ABC's template required for submission to ABC.
3.	BCA Part B1D4	All new glazing must comply with AS1288 (2021) and AS204 (2014).	To be confirmed.
4.2. Section C – Fire Resistance			
<u>Part C2 – Fire Resistance and Stability</u>			
1.	BCA Part C2D2	The Type of Construction is determined based on the following: - The building can be considered a large isolated building under Part C3D4 as it exceeds the floor area and volume limitations under Table C3D3 for a class 9b building. - On the plans provided the building is <u>single storey</u>. Provided the building is single storey the building can be considered a <u>Type C large, isolated building</u>.	Confirmation required from the fitout designers that the fitouts will be single storey.
2.	BCA Part C2D3	The base building is considered a single storey building however under Clause C2D3 the following is noted: - A plant area situated at the top of the building that contains only heating, ventilating or water tanks, or similar service and equipment is not required to be counted in the rise of storeys of the building. - If any raised platforms or similar structures are installed as part of the fitout and they have storage or an occupiable space underneath the platform, then the building would be considered a two-storey class 9b large isolated building and this report would need to be updated to permit Type B construction which would require non-combustible external walls and fire ratings to structure etc. - A mezzanine is not considered a storey if it has any area less than 200m², is less than 1/3 the floor area of the room	Confirmation required from the fitout designers that the fitouts will be single storey.

Item	Reference	Issue	Recommendation
		and has no enclosing walls. If more than 2 mezzanines are provided in the same room the area of both mezzanines must not exceed 200m ² .	
3.	BCA Specification S5C24	The building is considered a single storey Type C Construction large isolated building which is more than 3m from a fire source feature. It is understood the building will be separated into 2 separate fire compartments therefore the following fire rated construction is required: • Fire wall separating buildings and fire compartments 90/90/90 FRL. • External walls exposed to a separate fire compartment 60/60/60 FRL (in accordance with the FER Project 05104107 Rev B). • Switch room 2-hour fire separation from the building if located within the building. • Comms room 2-hour fire separation from the building.	Provide a fire performance plan for any fire rated construction required.
4.	BCA Part C2D10	As a class 9b single storey large, isolated building the requirements of this clause (all components of external walls to be non-combustible) is not applicable. It is noted that the elevations show materials such as Danpalon and Kingspan insulated panels which are not non-combustible and only permitted if the building is single storey and Type C Construction.	Recommend discussions with BAC to ensure combustible materials are acceptable under the design brief.
5.	BCA Part C2D11	Fire hazard properties of any material or assembly within the building must comply with Spec C2D11 for floor, wall or ceiling linings. Provide fire hazard properties test reports as follows: • AS/ISO 9239 test report for any new floor coverings • AS 5637.1 for any proposed wall and ceiling linings Provide test reports from a registered testing authority for all materials forming linings in the building to include the Kingspan insulated panels and Danpalon material shown on the sections if they will form internal linings.	Provide test reports from a registered testing authority for all materials forming linings in the building.
6.	BCA Part C2D12	Precast or tilt-panel concrete external walls must comply with Spec C12D12 when used in a building with a rise in storey of not more than 2.	To be confirmed on structural engineer's design certificate.
<u>Part C3 – Compartmentation and Separation</u>			
7.	BCA Part C3D3	Fire compartments within the building must not exceed the limits of Table C3D3, except as permitted by Part C3D4 of the BCA. The building has the following characteristics: • Total floor area - Approx. 10,715m² - to be confirmed • Total volume - Approx. 113,210m³ - to be confirmed As the volume would exceed 108,000m³ the building would require <u>smoke exhaust</u>. Therefore, it is understood the building will be separated into 2 separate buildings and fire compartments as follows: <u>Fire compartment 1 (Play Centre Area 51)</u> • Area 7,115m² • Volume 71,810m³	Provide fire performance plans to demonstrate fire rated construction to be provided.

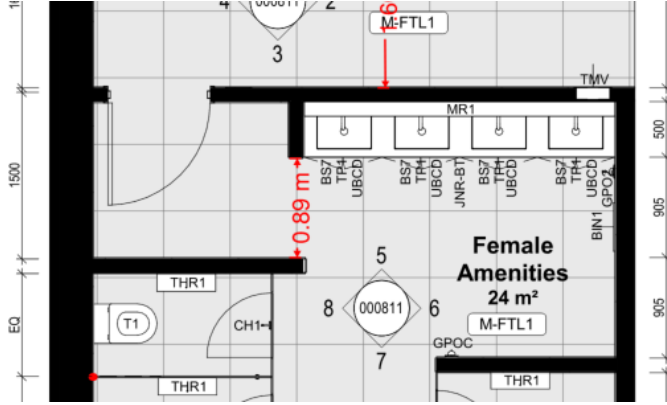
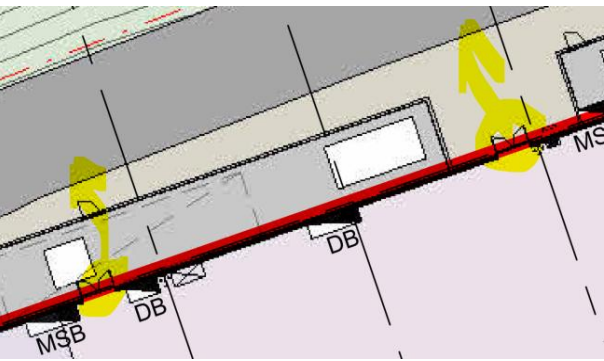
Item	Reference	Issue	Recommendation
		Fire compartment 2 (Game Over) - Area 3,600m² - Volume 41,400m³ If this fire separation is provided: - The building would be a large isolated building and require sprinkler protection and a vehicular access road. - However, the volume would be less than 108,000m³ and smoke exhaust would not be required. The fire engineering report FER Project 05104107 Rev B varies the extent of fire ratings required and requires a 4.5m long 90/90/90 fire rated wall to the Game Over tenancy that is exposed to the glazed foyer of the Area 51 tenancy. 	
8.	BCA Part C3D4	A building with fire compartments that exceed the limits in Table C3D3 must be considered a large, isolated building as permitted under C3D4 and the following is required: - A sprinkler system complying with Specification 17 must be provided throughout the building. - Vehicle perimeter access as per Part C3D5 (2)	Fire services engineer to provide sprinkler drawings and design certificate to ABC's template for submission to ABC.
9.	BCA Part C3D5	Fire brigade perimeter access must be achieved for a large isolated building. To comply, the vehicular access must: - Be capable of providing continuous access for emergency vehicles to enable travel in a forward direction around the entire building - Have a minimum unobstructed width of 6m with no part of its furthest boundary more than 18m from the building - Provide reasonable pedestrian access from the vehicular access to the building - Have a loadbearing capacity and unobstructed height to permit operation and passage of fire brigade vehicles. - Be wholly within the allotment except that a public road complying with the above may serve as the vehicular access or part thereof. Following a review of plans the following is noted: - The access road must be able to withstand a 30-tonne truck and 10kg/m² puncture load – to be confirmed on the Structural engineers Design Certificate. - The perimeter access road furthest boundary is up to 52m away from the building in lieu of max 18m from the building – this must be documented as a fire engineered performance	Structural engineer to confirm compliance with this clause on their Design Certificate. Fire engineered performance solution required to permit the furthest boundary of the perimeter access road to be up to 52m away from the building in lieu of max 18m from the building. This must be discussed with Airport Building Control, and airports' fire brigade (ARFF).

Item	Reference	Issue	Recommendation
		solution to be reviewed by the ABC. The road appears to be less than the required 6m wide in parts – please provide detailed dimensioned drawings showing the road at 6m wide at all points around the building.	
10.	BCA Part C3D8	A fire wall provided in accordance with C3D8(2) may be provided to separate buildings for the application of the BCA Sections C, D and E. It is understood a 90/90/90 fire wall will be provided to separate the building into 2 separate buildings limiting the size of fire compartments to not greater 18,000m² in area and 108,000m³ in volume (to avoid the requirement to provide smoke exhaust to the building). The following applies to the fire wall: - The fire wall must achieve a 90/90/90 FRL. - Service openings in the fire wall must be protected in accordance with Clause C4D15 - The fire wall must continue to the underside of the roof covering. - Building elements, other than roof battens with dimensions of 75 mm x 50 mm or less or sarking-type material, must not pass through or cross the fire wall unless the required fire-resisting performance of the fire wall is maintained. If an insulated roof sheet is proposed as the roof covering it must not cross the fire wall as it would be combustible – please provide a detail of the junction of the fire wall and roof covering as highlighted in yellow below to show the fire wall extending above the roof or other solution to comply. 	Provide a detail of the junction between the fire wall and roof covering to demonstrate that the fire resisting performance of the fire wall is maintained.
11.	BCA Part C3D13	Equipment considered an explosion risk by BCA are required to be 2-hour fire separated from the remainder of the building and have any openings protected in accordance with Part C4 of the NCC. Such equipment includes batteries, central smoke control plant emergency generators, etc. Confirmation required if the building will contain a comms room with a battery system having a total voltage of 12 volts or more and a storage capacity of 200 kWh or more.	MSB boards are shown as fire separated from the building – confirm the FRL achieved to the walls and ceiling will be 2-hours on plans.
12.	BCA	Electricity supply systems which sustain emergency equipment is	MSB boards are shown as fire

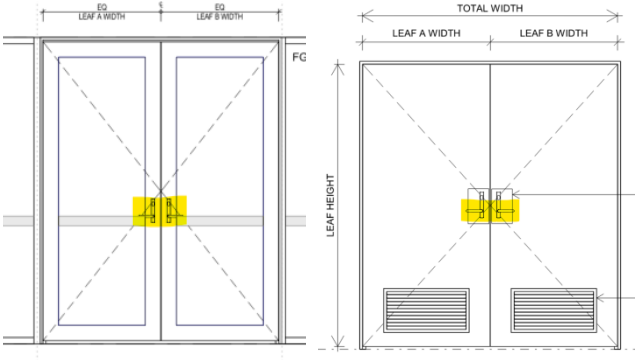
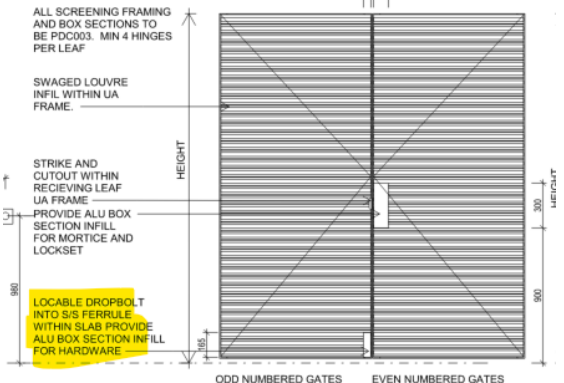
Item	Reference	Issue	Recommendation
	Part C3D14	to be 2-hour separated from the remainder of the building with any openings protected in accordance with Part C4 of the NCC. Examples of such equipment includes electrical substations, main switchboards, electrical conductors and plant for smoke exhaust.	separated from the building - confirm the FRL achieved will be 2-hours on plans.
Part C4 – Protection of Openings			
13.	BCA Part C4D3	Any openings in an external wall required to have an FRL must be protected if the distance between the opening and a fire source feature is less than: 3m from side and rear allotment boundaries 6m from the far boundary of a road 6m from another building that is not a Class 10. The building is not exposed to a fire source feature and protection of openings is not required under this clause.	Note
14.	BCA Part C4D4	Where openings in the external walls of separate fire compartments are exposed to one another, they must be protected in accordance with Part C4. It is understood the building will be separated into 2 separate buildings and fire compartments with a 90/90/90 fire wall. External walls of separate fire compartments are exposed as follows: - Entry foyer - external wall of the Play Centre Area 51 is exposed and less than 6m from the external wall of the Game Over tenancy - Rear façade - external wall of the rear of the Pay Centre Area 51 is exposed at 90 degrees to the external wall of the Game Over tenancy The following would be required: - Protection of openings would be required in accordance with Part C4 with 60/60/60 fire rated walls and -/60/30 fire rated doors provided on both sides of the fire wall The fire engineering report FER Project 05104107 Rev B varies the extent of fire ratings required above and requires a 4.5m long 90/90/90 fire rated wall to the Game Over tenancy where it is exposed to the glazed foyer of the Area 51 tenancy.	Fire engineered performance solution required to rationalize exposure of openings in external walls exposed to a separate fire compartment. This must be discussed with Airport Building Control, and airports' fire brigade (ARFF).



Item	Reference	Issue	Recommendation												
15.	BCA C4D15	Penetrations through fire rated elements for services must be fire sealed to maintain the integrity and insulation requirements of the fire rated element. The method of protection is to be a tested system in accordance with AS 1530.4 (2014) with details of the system provided to the certifier for review.	Note												
4.3. Section D – Access & Egress															
Part D2 – Provision for Escape															
1.	BCA Part D2D3	As a 9b building accommodating more than 50 persons, 2 exits are required to all areas.	Note												
2.	BCA Part D2D5	Travel distances to an exit must meet the following requirements— 20m to a single exit or to a point of choice to 2 exits, with total travel to one of those exits being no more than 40m 30m to a single exit on the ground floor in the class 6 part only. The base building (open plan design) is capable of compliant travel distances however, in order to 'future proof' the tenancy fitout layouts the fire engineer has justified the following travel distances: 30m in lieu of max 20m to a point of choice between alternative exits 60m in lieu of max 40m to an exit	An additional exit door required or extended travel to be addressed as a fire engineered solution Extended travel distances to be justified by the fire engineer. This must be discussed with Airport Building Control, and airports' fire brigade (ARFF).												
3.	BCA Part D2D6	Alternative exits must be located: - Not less than 9m apart - Not more than 60m apart. - Must not converge to less than 6m apart. The base building (open plan design) has extended distance between exits of approx. 70m between alternative exits in lieu of max 60m as an open plan layout and an additional exit door required or fire engineered performance solution required. In order to 'future proof' the fitout layouts the fire engineer has justified the following distance between alternative exits: 90m between alternative exits in lieu of max 60m.	An additional exit door required or extended travel to be addressed as a fire engineered solution This must be discussed with Airport Building Control, and airports' fire brigade (ARFF).												
4.	BCA Part D2D7	All exit paths must achieve a minimum height of 2000mm (except 1980mm is permitted at doorways).	Note												
5.	BCA Part D2D8	All exit paths must maintain a minimum of 1000mm clear width. The following egress width is required in the building: <table border="1"> <thead> <tr> <th>Area</th><th>Occupancy</th><th>Egress width required</th><th>Egress width provided</th></tr> </thead> <tbody> <tr> <td>Play Centre</td><td>Max 525 persons</td><td>4.5m</td><td>Approx. 15m provided.</td></tr> <tr> <td>Game Over</td><td>Max 500 persons (internal)</td><td>4m</td><td>Approx. 8.5m+ provided</td></tr> </tbody> </table> The following is noted: The above calculations assume double exit doors will provide a minimum of 1.5m clear egress width.	Area	Occupancy	Egress width required	Egress width provided	Play Centre	Max 525 persons	4.5m	Approx. 15m provided.	Game Over	Max 500 persons (internal)	4m	Approx. 8.5m+ provided	Note End user to confirm maximum expected population at any given time.
Area	Occupancy	Egress width required	Egress width provided												
Play Centre	Max 525 persons	4.5m	Approx. 15m provided.												
Game Over	Max 500 persons (internal)	4m	Approx. 8.5m+ provided												

Item	Reference	Issue	Recommendation
		Please confirm all paths to an exist are a minimum of 1m wide. It appears there may be a pinch point less than 1m wide in the female Game over amenities.  The floor plan shows the 'Female Amenities' area (24 m²) with various rooms and exits. A red arrow points to a narrow section of the exit path labeled '0.89 m'. Other labels include 'MFTL1', 'MR1', 'BS7', 'TR4', 'UBCD', 'JNR-BT', 'BIN1', 'GPOC', 'THR1', 'CH1', 'T1', and 'EQ'.	
6.	BCA Part D2D9	All exit doors must have a minimum width of 750mm clear width. Note - where a door is required to be accessible the door must achieve 850mm clear width.	Note
7.	BCA Part D2D10	The unobstructed width of a required exit must not diminish in the direction of travel to a road or open space.	Drawings to show a path from exit doors to the road
8.	BCA Part D2D15	Exits must not be blocked at the point of discharge and where necessary, suitable barriers must be provided to prevent vehicles from blocking the exits. - Bollards to be provided to protect external doors where they could be blocked by vehicles. - The gradient of an exit path to the road (open space) must not exceed a gradient of 1:8 – confirm the gradient of exit paths on the plans. Please confirm the following: - Confirm how a compliant 1m wide egress path will be provided from the exit door through the plant area. - Show a path linking the exit doors to the road and confirm gradients. - Provide a detail of thresholds to egress doors.  The site plan shows the building footprint, exit doors, and proposed bollard locations (yellow circles) along the exit path to the road. Labels include 'MSB', 'DB', and 'MS'.	Provide bollards to prevent exits from being blocked by vehicles where applicable. Confirm egress paths between exit doors and the road on plans.

Item	Reference	Issue	Recommendation															
9.	BCA Part D2D18	The number of persons accommodated within the building must be calculated by either: - Table D2D18 persons per square metre or - Occupancy to be noted on plans as advised by the client/end user. Based on confirmation received from the client the following occupancy is calculated: <table><tr><th>Area</th><th>Area per person</th><th>Occupancy</th></tr><tr><td colspan="3">Play Centre</td></tr><tr><td>Play area</td><td>Total 525 persons – as advised</td><td>525 persons</td></tr><tr><td colspan="3">Game Over</td></tr><tr><td>Game over</td><td>Total 500 persons – as advised</td><td>500 persons</td></tr></table>	Area	Area per person	Occupancy	Play Centre			Play area	Total 525 persons – as advised	525 persons	Game Over			Game over	Total 500 persons – as advised	500 persons	Note
Area	Area per person	Occupancy																
Play Centre																		
Play area	Total 525 persons – as advised	525 persons																
Game Over																		
Game over	Total 500 persons – as advised	500 persons																
<u>Part D3 – Construction of Exits</u>																		
10.	BCA Part D3D8	All electrical distribution boards/meters and central telecommunications boards may be located within a path of travel to an exit. However, the system must be protected within a cabinet smoke sealed and non-combustible construction.	Note															
11.	BCA Part D3D16	In a building required to be accessible for people with disabilities, a doorway that opens to a road or open space must be provided with a threshold ramp or step ramp in accordance with AS1428.1 (2021) if a change in level applies. In other cases – a step may be permitted if the doorway opens to a road or open space and the door sill is not more than 190mm above the finished floor level of the ground to which the doorway opens.	Please confirm detail of external thresholds.															
12.	BCA Part D3D17	A balustrade must be provided where a fall greater than 1m is possible along a trafficable path. Where a balustrade is required, the following applies: - Minimum height of 1000mm. - It is recommended this be increased to a 1050mm to allow for construction tolerances.	Confirm any level difference of over 1m on trafficable paths.															
13.	BCA Part D3D24	The following applies to doors serving a required exit: - Must not be fitted with a sliding door unless the door leads to a road or open space and can be manually opened with a force not exceeding 110N. - If power-operated, the door must be capable of being manually opened with a force not exceeding 110N. - Automatic doors if they lead directly to a road or open space, it must open automatically on power failure or on activation of a fire or smoke alarm. - A power-operated door in a path of travel to an exit must be capable of being opened manually with a force not exceeding 110N if power failure.	Provide a note on the plans to applicable doors on door schedule. Electrical services engineer to provide drawings and design certificate to ABC's template for submission to ABC as applicable.															
14.	BCA Part D3D25	Exit doors must swing in the direction of egress. To be reviewed as the design progresses.	Note															
15.	BCA	A door in a required exit, or a door in a path of travel to an exit	Provide a door															

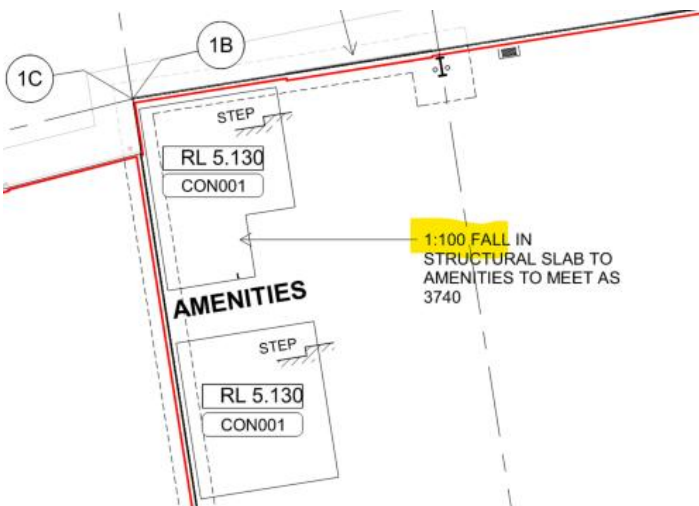
Item	Reference	Issue	Recommendation
	Part D3D26	must be readily openable without a key from the side that faces a person seeking egress and the following applies: - The building is a class 9b building accommodating more than 100 persons therefore all doors must be openable with a push action such as push bars. - The push bars or buttons must be located on the door. - The push bar or button must be located between 900mm and 1.2m high. Plans show exit door with lever handles which would not comply with this clause as a 9b building with over 100 persons – and plans to be updated to show push buttons or push bars to all exit doors.  The G01 gate is an egress gate in some locations and cannot have drop bolts – confirm where drop bolts will be located and egress door must not have drop bolts.  G01 - ALUMINIUM LOUVRE GATE	schedule showing push bars or buttons to all EXIT doors.
Part D4 – Access for People with Disabilities			
16.	Part D4D2 & D4D3	People with Disabilities (PWD) access is required to and within all areas normally used by the occupants and as follows: - From the main points of a pedestrian entry at the allotment boundary to the building main entry. - From wheelchair accessible car parking spaces to the building main entry. - Through the main entrance and 50% of all other building entries. - From another accessible building on the allotment connected	Provide a site plan to show all levels and gradients from allotment pedestrian boundaries and accessible carparking spaces up to and into the building.

Item	Reference	Issue	Recommendation
		by a pedestrian link.	
17.	Part D4D4	Please see Knisco Inclusive access report/markups for a detailed review of access requirements and the following is highlighted: - Access must be provided to and within all areas of the building in accordance with AS1428.1 (2021). - Provide detailed drawings of all accessible sanitary facilities for review - Provide a door schedule to confirm compliance with AS1428.1 (2021). - Provide details of accessible door thresholds to outdoor areas. <u>General comments:</u> - End of corridors must have a turning space 1540mm x 2070mm setback no further than 2m from end of corridor. - A passing space (2m x 1.8m) must be provided at 20m intervals where there is no direct line of sight. - All doorways must achieve a minimum 850mm clear opening and for double doors the active leaf must achieve 850mm clear width. - All swing doors must have a lever handle that turns in at the end and all sliding doors have a "D" type door handle. - All accessible doors must have a 50mm wide 30% luminance contrast between the door and the jamb, architrave, wall, etc. - All glazing that is capable of being mistaken for a doorway must be provided with motifs/decals.	Please see Knisco Inclusive Access Report KIA-Q258334 for an in-depth review disability access to the building. Bio-fold doors to external areas of Game Over requires a dedicated wheelchair accessible door.
18.	Part D4D5	Certain areas of a building are not required to be provided with disability access if: - Access would be inappropriate because of the particular purpose for which the area is used. - Access to that area would pose a health or safety risk for a person with a disability. Confirm any area where disability access will not be provided and this concession to be applied.	Any concessions must be granted by Airport Building Control.
19.	Part D4D6	Accessible PWD car parking spaces are required for a class 9b building at a rate of 1 space for every 50 spaces provided. The plans provided show approx. 90 car parking spaces provided which would require 2 accessible PWD car parking spaces Detailed drawings required to show the dimensions of the space, gradient of the space, hatched circulation space, bollard etc. in accordance with AS 2890.6 (2009) Confirm any kerb ramps etc. to be provided between the accessible spaces and the footpath. Please show the hatch circulation space, bollard and signage to the accessible PWD car parking spaces. Please see Knisco Inclusive Access Report KIA-Q258334 for an in-depth review of accessible carparking spaces.	Provide detailed drawings of accessible car parking spaces.
20.	Part D4D7	Accessible facilities will require: All toilets (male/female/disabled/ambulant) must be provided	Provide details of all braille signage required in the

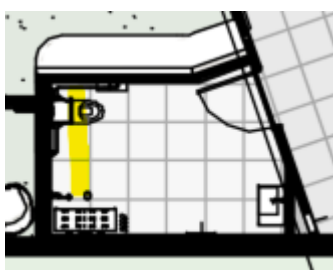
Item	Reference	Issue	Recommendation
		with braille and tactile signage to the door leading into each sanitary facility. - The signage must be located with tactile lettering between 1250mm and 1350mm. - The location of the sign must be provided on the latch side of the wall between 50-300mm away from the architrave. If this is not possible, signage may be provided to the door. - For signs to ambulant accessible compartments, this signage <u>must</u> be provided on the door to the facility. Exit door braille and tactile signage must be provided to indicate the word "EXIT" and which level the door is located on. Please see Knisco Inclusive Access Report KIA-Q258334 for an in-depth review of accessible sanitary facilities.	building.
21.	Part D4D8	A hearing augmentation system must be provided where an inbuilt amplification system, other than one used only for emergency warning, is installed in: - A room in a Class 9b building; or - A ticket office, teller's booth, reception area or the like, where the public is screened from the service provider.	Provide confirmation if a hearing augmentation system will be provided to the building and if applicable details to be provided.
22.	Part D4D9	Tactile indicators must be provided in accordance with AS1428.4.1 (2021) to warn people who are vision impaired that they are approaching: - Any overhead obstruction less than 2000mm in height (e.g. the underside of a stair) - An access way meeting a vehicular way if there is no kerb or kerb ramp provided at that point. Provide drawings showing tactile indicators where paths meet the road at the same level. Confirm tactiles to paths that meet the road at the same level such as the path adjacent to the bike racks and foots paths generally if no kerb is provided to the footpath. 	Provide a drawing showing with tactile indicators where required.

Item	Reference	Issue	Recommendation
4.4. Section E – Services & Equipment			
<u>Part E1 – Fire Fighting Equipment</u>			
1.	BCA Part E1D2	As the building is over 500m² a fire hydrant system is required in accordance with AS2419.1 (2021) and the following applies: • Provide a flow and pressure test for the site. • Hydraulic consultant to provide fire hydrant coverage drawings. Please confirm if the following fire engineered performance solutions will be required: • It is understood that fire separation will be provided in the building (in accordance with C3D8 (2), therefore the volume of the building will not exceed 108,000m³ and application of the fire hydrant system via a performance solution will not be required (AS 2419.1 2021 Section 1, 1.1 scope excludes buildings with a volume over 108,000m³ in volume) • If internal hydrants are installed/required more than 4m from an exit they must be documented as a fire engineered performance solution. • The location of the booster assembly is not in the façade of the building containing the principal pedestrian entrance.	Fire services engineer to provide drawings and design certificate to ABC's template for submission to ABC. Fire engineered performance solution required to vary the requirements of AS 2419.1 (2021). This must be discussed with Airport Building Control, and airports' fire brigade (ARFF).
2.	BCA Part E1D3	A fire hose reel system is required to serve the building in accordance with AS2441 (2005), and the following applies: • Fire hose reels must be located within 4m of an exit. • Where coverage is not achieved by fire hose reels within 4m of an exit additional fire hose reels may be located on the paths of travel to an exit.	Fire services engineer to provide drawings and design certificate to ABC's template for submission to ABC.
3.	BCA Part E1D4	As per Part C3D4 & Part E1D12 of the BCA, a class 9b large isolated building must be provided with an AS 2118.1 (2017) sprinkler system throughout complying with Specification 17 of the NCC. A fire engineered performance solution to be provided to address: • Omission of sprinklers protection below play equipment exceeding 800 mm width within the Area 51, in lieu of complying with AS 2118.1, which requires additional sprinklers under obstruction of this width.	Fire services engineer to provide drawings and design certificate to ABC's template for submission to ABC. Fire engineered performance solution must be discussed with Airport Building Control, and airports' fire brigade (ARFF).
4.	BCA Part E1D14	Fire extinguishers must be selected, located and distributed in accordance with AS 2444 (2001) and must be provided to protect: • To cover Class AE/E fire risks associated with emergency services switchboards. • To cover Class F fire risks associated with cooking oils & fats in kitchens Pictorial signage must be located 2m above FFL. If the extinguisher is located within a cupboard additional signage must be provided to the door stating FIRE EXTINGUISHER 50mm high and on a contrasting background.	Location of fire extinguishers to be noted on plans where required.

Item	Reference	Issue	Recommendation
<u>Part E2 – Smoke Hazard Management</u>			
5.	BCA Part E2D10	The following smoke hazard management systems would be required: - AS 2118.1 (2017) sprinkler system - AS 1670.1 (2018) occupancy warning system - AS 1670.3 (2018) monitoring by fire brigade. - Location of the FIP to be confirmed by fire services contractor (currently appears to be located in the pump room in lieu of the building main entry foyer). The following is noted: <u>Clause E2D10</u> - Fire separation will be provided to the building in accordance with Clause C3D8 (2) to permit the 2 tenancies to be considered separate buildings each not greater than 18,000m² in area and 108,000m³ in volume thus the provision of smoke exhaust is not triggered. <u>Clause E2D14</u> - The class 6 café parts are less than 2000m² in area thus the provision of smoke exhaust is not triggered. <u>Clause E2D20</u> - The Play Centre Area 51 and Game Over buildings are class 9b (Area 51 has an area greater than 5000m²) however it is considered they have a use similar to a sports hall or gymnasium or the like and thus the requirements of this clause are not triggered.	Fire services engineer to provide drawings and design certificate to ABC's template for submission to ABC. This must be discussed with Airport Building Control, and airports' fire brigade (ARFF).
<u>Part E4 – Visibility in an Emergency, Exit Signs and Warning Systems</u>			
6.	BCA Part E4D2 to E4D5	Emergency lighting and exit signage system is required throughout the building in accordance with Part E4 of the BCA and AS2293.1.	Electrical engineer to provide drawings and design certificate to ABC's template for submission to ABC.
7.	BCA Part E4D9	The building is a class 9b building however is not considered to be a public hall having a floor area of over 10000m ² therefore an EWIS system is not required.	Note
4.5. Part F – Health & Amenity			
<u>Part F1 – Damp and Weatherproofing</u>			
1.	BCA Part F1D3	Stormwater drainage is to be in accordance with AS3500.3.	Hydraulics engineer to provide drawings and design certificate to ABC's template for submission to ABC.
2.	BCA Part F1D5	Confirmation required on how external door thresholds will be waterproofed with details required to demonstrates threshold ramps, grated drains, water stop angles etc.	Provide detail of thresholds for review.
<u>Part F2 - Wet areas and overflow protection</u>			
3.	BCA Part F2D2	Water proofing of wet areas is to be designed in accordance with Specification 26 and AS 3740 (2021) and manufacturers specifications.	Provide note on drawings to confirm.

Item	Reference	Issue	Recommendation
4.	BCA Part F2D3	Where a urinal is installed a floor waste must be provided and the wall, floor and wall/floor junction must be impervious. <i>Floor wastes to be shown on plans to showers and urinals.</i>	Provide note on drawings to confirm.
5.	BCA Part F2D4	Where a floor waste is installed (required adjacent to the urinal and to shower areas) in wet areas the floor must fall to the outlet with a gradient of between 1:50 to 1:80. <i>Currently the drawings note falls of 1:100 to the structural slab in accordance with AS 3740 in wet areas. Please confirm the fall in the floor plain will be between 1:50 to 1:80.</i> <i>Drawings show a set down of 70mm to wet areas – please confirm that level access will be provided to sanitary facilities with no step to doors and the step down is provided to allow for falls and tiles etc.</i> 	Provide note gradients of floors drawings to confirm.
Part F3 - Roof and wall cladding			
6.	BCA Part F3D2 & F3D3	Roof coverings to comply as follows: • Metal roof sheeting – AS1562.1 • Sarking must comply with AS/NZS 4200 Parts 1 & 2.	Provide note on drawings to confirm.
7.	BCA Part F3D5	External wall cladding must comply with one or a combination of the following: • Masonry, including masonry veneer, unreinforced and reinforced masonry: AS 3700 (2018) • Autoclaved aerated concrete: AS 5146.3 (2018). • Metal wall cladding: AS 1562.1. (2018) A performance Solution must be documented to demonstrate how the external wall will be weatherproof where the following occurs: • A combination of the above materials are installed in the external walls. • Materials not listed above are installed as part of the external cladding of the external walls.	Given the external walls include materials such as Kingspan insulated panels and Danpalon a performance solution must be documented to detail the weatherproofing of external walls in the building.

Item	Reference	Issue	Recommendation																																																																																																																													
Part F4 – Sanitary and Other Facilities																																																																																																																																
8.	Part F4D3 & F4D4	Sanitary facilities must be provided in the building based on the building's occupancy. In accordance with advice on max occupancy numbers provided the following sanitary facilities would be required: Play Centre Area 51 - Numbers are calculated based on an occupancy of 525 patrons, 25 staff, 60 dining patrons and 440 patrons – as advised by the client. - The PWD accessible facility has been counted in the numbers as 1 male and 1 female facility. - In a class 9b building staff and patrons can share facilities. <table><tr><th colspan="5">Play Centre Area 51</th></tr><tr><th></th><th>Pan</th><th>Urinal</th><th>Basin</th><th>Comment</th></tr><tr><td>PWD toilet</td><td>1</td><td></td><td>1</td><td>Provided.</td></tr><tr><th colspan="5">Play Centre Area 51 Female facilities</th></tr><tr><td>13 female staff</td><td>1</td><td></td><td>1</td><td></td></tr><tr><td>220 female patrons</td><td>5</td><td></td><td>3</td><td></td></tr><tr><td>30 female café diners</td><td>2</td><td></td><td>1</td><td></td></tr><tr><td>Total female facilities</td><td>7</td><td></td><td>4</td><td>Provided.</td></tr><tr><th colspan="5">Play Centre Area 51 Male facilities</th></tr><tr><td>13 male staff</td><td>1</td><td>1</td><td>1</td><td></td></tr><tr><td>220 male patrons</td><td>1</td><td>3</td><td>2</td><td></td></tr><tr><td>30 male café diners</td><td>1</td><td>1</td><td>1</td><td></td></tr><tr><td>Total male facilities</td><td>2</td><td>5</td><td>3</td><td>Provided.</td></tr></table> Game Over tenancy - Numbers are calculated based on an occupancy of 500 patrons, 20 staff, 60 dining patrons and 420 patrons at any given time – as advised by the client. - The PWD accessible facility has been counted in the numbers as 1 male and 1 female facility. - In a class 9b building staff and patrons can share facilities. <table><tr><th colspan="5">Game over tenancy</th></tr><tr><th></th><th>Pan</th><th>Urinal</th><th>Basin</th><th>Comment</th></tr><tr><td>PWD toilet</td><td>1</td><td>-</td><td>1</td><td>Provided.</td></tr><tr><th colspan="5">Game over Female facilities</th></tr><tr><td>10 female staff</td><td>1</td><td></td><td>1</td><td></td></tr><tr><td>210 female patrons</td><td>5</td><td></td><td>3</td><td></td></tr><tr><td>30 female café diners</td><td>2</td><td></td><td>1</td><td></td></tr><tr><td>Total female facilities</td><td>7</td><td></td><td>4</td><td>Provided.</td></tr><tr><th colspan="5">Game Over Male facilities</th></tr><tr><td>10 male staff</td><td>1</td><td></td><td>1</td><td></td></tr><tr><td>210 male patrons</td><td>1</td><td>3</td><td>2</td><td></td></tr><tr><td>30 male café diners</td><td>1</td><td>1</td><td>1</td><td></td></tr></table>	Play Centre Area 51						Pan	Urinal	Basin	Comment	PWD toilet	1		1	Provided.	Play Centre Area 51 Female facilities					13 female staff	1		1		220 female patrons	5		3		30 female café diners	2		1		Total female facilities	7		4	Provided.	Play Centre Area 51 Male facilities					13 male staff	1	1	1		220 male patrons	1	3	2		30 male café diners	1	1	1		Total male facilities	2	5	3	Provided.	Game over tenancy						Pan	Urinal	Basin	Comment	PWD toilet	1	-	1	Provided.	Game over Female facilities					10 female staff	1		1		210 female patrons	5		3		30 female café diners	2		1		Total female facilities	7		4	Provided.	Game Over Male facilities					10 male staff	1		1		210 male patrons	1	3	2		30 male café diners	1	1	1		Occupancy numbers for the building to be confirmed. Additional sanitary facilities are required in the building to comply.
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Item	Reference	Issue	Recommendation					
		<table><tr><td>Total male facilities</td><td>2</td><td>4</td><td>3</td><td>Provided.</td></tr></table> It is noted that on the current plans the sanitary facility basins are located in a communal area in lieu of being dedicated male and female basins which would need to be documented as a performance solution.	Total male facilities	2	4	3	Provided.	
Total male facilities	2	4	3	Provided.				
9.	Part F4D5 & F4D6	In a building required to be accessible: - An accessible PWD unisex sanitary compartments must be provided to both the Play centre facilities and the Game Over tenancy facilities – provided. - At each bank of toilets, in addition to an accessible unisex sanitary compartment, a male ambulant toilet and a female ambulant toilet must be provided in accordance with AS1428.1 (2021) - Circulation spaces, fixtures and fittings of all accessible sanitary facilities provided must comply with AS1428.1. (2021). - If more than 1 unisex accessible sanitary compartment is required, left and right-hand transfer facilities must be provided. Please see Knisco Inclusive Access Report KIA-Q258334 for an in-depth review of sanitary facilities and the following is highlighted: Game Over facilities - The baby change table is located within the toilet and sink circulation and must be omitted or re-positioned. - Clothes hooks in the PWD accessible toilet must be between 1200 and 1350mm high and within 400mm and 600mm of the seat to be adjusted on plans. General - Confirm the 1:60 to 1:80 gradient to the accessible PWD shower. Confirm the set out of the accessible shower floor waste in accordance with AS 1428.1. - It is noted that on current plans both PWD toilets are left hand transfer, and plans to be amended to show one left hand transfer (LH) and the other right-hand transfer (RH). 	Please provide detailed drawing of accessible sanitary compartment and ambulant accessible sanitary compartment for further assessment					
10.	Part F4D8	Where cubicle door frames are within 1.2m of the pan either of the following must be provided: - Outward opening door; or - Sliding door; or - Have lift-off hinges provided.	Note					

Item	Reference	Issue	Recommendation
11.	Part F4D12	A class 9b sports venue requires adult change facilities if it has a design occupancy of greater than 35,000 spectators. It appears that the Play Centre and the Game Over tenancy would not accommodate more than 35,000 spectators and adult change facilities would not be required.	Please confirm the occupancy of the building.
<u>Part F5 - Room heights</u>			
12.	BCA Part F3.1	Minimum height of ceilings within the buildings must be as follows: - Habitable rooms – 2.7m - Bathrooms and associated corridors and the like – 2.1m - Commercial kitchen for café – 2.4m The building is a class 9b assembly building therefore all areas containing more than 100 persons and associated corridors must have a ceiling height of not less than 2.7m high.	Ceiling plan required for review.
<u>Part F6 - Light and ventilation</u>			
13.	BCA Part F6D5	Artificial lighting must be provided to all room that are frequently occupied, all spaces required to be accessible, all corridors, circulation spaces and paths of egress in accordance with AS/NZS1680.0 (2009).	Electrical engineer to provide drawings and design certificate to ABC's template for submission to ABC.
14.	BCA Part F6D6 & F6D7	Ventilation of habitable rooms must be achieved through either: - Natural ventilation – 5% of floor area of room; or - Mechanical ventilation in accordance with AS1668.2 (2012) and AS3666.1(2011).	Mechanical engineer to provide drawings and design certificate to ABC's template for submission to ABC.
15.	BCA Part F6D9	Sanitary facilities cannot open directly into a room such as a workplace occupied by more than one person unless an airlock or mechanical ventilation and screening of the doorway is provided. <i>On current plans sanitary facilities open into the Game Over and Area 51 tenancies without an air lock therefore the doors to the facilities must be screened from view.</i>	To be confirmed as the design progresses.
16.	BCA Part F6D12	Commercial kitchens must be provided with a kitchen exhaust hood complying with AS1668.1 and AS1668.2. if the following will be installed: - If any cooking apparatus within this kitchen has a total maximum electrical input exceeding 8kW or a total gas power input exceeding 29MJ/h; or - The total maximum power input to more than one apparatus exceeds 0.5kW electrical power or 1.8MJ gas per m² of the floor area of the kitchen. Confirmation required if the base building work will include cooking facilities in the cafés that would include facilities listed above. If provided confirm how kitchen exhaust will discharge from the building and Design Certificate required by mechanical consultant.	Mechanical engineer to provide drawings and design certificate to ABC's template for submission to ABC.

Item	Reference	Issue	Recommendation
Section G – Ancillary Provisions			
<u>Part G1 – Minor Structures and Components</u>			
17.	BCA Part G1D3	A cool room provided in a building must be provided with: • An internal light controlled only by a switch which is located adjacent the entrance door of the cold room • An indicator lamp located outside the cold room which is illuminated when the internal light is switched on • An alarm that is located outside the cold room but controllable only from the inside and capable of achieving a sound pressure level of 90dB(A) • A door with a clear width of not less than 600mm and a height of not less than 1.5m capable of being opened from the inside without the use of a key Confirmation required if the base building work will include cool rooms in the café that would be required to comply with the above.	Confirm if any cold room to be included in the base building work.
<u>Section J – Energy Efficiency</u>			
18.	BCA Part J1 – J7	The proposed building design needs to comply with Section J Energy Efficiency Parts: • J4 – Building Fabric. • J5 – Building Sealing. • J6 – Air-conditioning and ventilation systems • J7 – Artificial lighting and power • J8 – Hot water supply • J9 – Facilities for energy monitoring In order to demonstrate compliance, the following is required – • For Parts J4 & J5 - An energy efficiency assessment is to be carried by a suitable qualified energy efficiency consultant. • For Parts J6, J7, J8 & J9 – must be designed and certified by the relevant services engineer or consultant on their Design Certificates.	Energy report required. Consultant Design Certificates to confirm compliance.
19.	BCA Part J9D3	Building with a floor area over 2,500m ² must have energy meters configured to enable individual time of use energy data recording of the building services including renewable.	Electrical Design Certificate to confirm compliance
20.	BCA Part J9D4	Design Certificate by Electrical consultant to confirm compliance with Clause J9D4 (1 & 2) in relation to the provision of electrical distribution boards for future installation of electrical vehicular charging.	Electrical Design Certificate to confirm compliance
21.	BCA Part J9D5	• The electrical consultants Design Certificate must confirm compliance Clause J9D5 Facilities for solar Photovoltaic and battery systems. • Architectural drawings must demonstrate that at least 20% of the roof area of a building has been left clear for the installation of solar photovoltaic panels – complies.	Electrical Design Certificate to confirm compliance

Please refer attached information from Airport Building Control in regard to the information and forms required to be submitted with the Building Approval Application.

The site plan illustrates a proposed development with several key components:

- AREA 51:** Building 340, 7,115sqm, including amenities, cafe/store, excluding outdoor screened seating & outdoor dining. Assumed site area 10,662.93 sqm.
- GAME OVER:** Tenant 2 Building 345, 10,662.93 sqm, including outdoor screened seating & outdoor dining. Assumed site area 500pax.
- ACTIVATION:** Indicative end of trip facilities, 31 bike racks, 30m.
- Service Roads:** Multiple service roads are shown, including one labeled "BATTERED GRADE TO NATURAL GROUND".
- Parking:** Short term parking (x 6 carriageway) and 30 cars @ 2500m (including 2 x PWD) are indicated.
- Dimensions:** Various dimensions are provided in meters (m) and feet (ft), such as 40.20m, 40.31m, 40.41m, 40.51m, 40.61m, 40.71m, 40.81m, 40.91m, 41.01m, 41.11m, 41.21m, 41.31m, 41.41m, 41.51m, 41.61m, 41.71m, 41.81m, 41.91m, 42.01m, 42.11m, 42.21m, 42.31m, 42.41m, 42.51m, 42.61m, 42.71m, 42.81m, 42.91m, 43.01m, 43.11m, 43.21m, 43.31m, 43.41m, 43.51m, 43.61m, 43.71m, 43.81m, 43.91m, 44.01m, 44.11m, 44.21m, 44.31m, 44.41m, 44.51m, 44.61m, 44.71m, 44.81m, 44.91m, 45.01m, 45.11m, 45.21m, 45.31m, 45.41m, 45.51m, 45.61m, 45.71m, 45.81m, 45.91m, 46.01m, 46.11m, 46.21m, 46.31m, 46.41m, 46.51m, 46.61m, 46.71m, 46.81m, 46.91m, 47.01m, 47.11m, 47.21m, 47.31m, 47.41m, 47.51m, 47.61m, 47.71m, 47.81m, 47.91m, 48.01m, 48.11m, 48.21m, 48.31m, 48.41m, 48.51m, 48.61m, 48.71m, 48.81m, 48.91m, 49.01m, 49.11m, 49.21m, 49.31m, 49.41m, 49.51m, 49.61m, 49.71m, 49.81m, 49.91m, 50.01m, 50.11m, 50.21m, 50.31m, 50.41m, 50.51m, 50.61m, 50.71m, 50.81m, 50.91m, 51.01m, 51.11m, 51.21m, 51.31m, 51.41m, 51.51m, 51.61m, 51.71m, 51.81m, 51.91m, 52.01m, 52.11m, 52.21m, 52.31m, 52.41m, 52.51m, 52.61m, 52.71m, 52.81m, 52.91m, 53.01m, 53.11m, 53.21m, 53.31m, 53.41m, 53.51m, 53.61m, 53.71m, 53.81m, 53.91m, 54.01m, 54.11m, 54.21m, 54.31m, 54.41m, 54.51m, 54.61m, 54.71m, 54.81m, 54.91m, 55.01m, 55.11m, 55.21m, 55.31m, 55.41m, 55.51m, 55.61m, 55.71m, 55.81m, 55.91m, 56.01m, 56.11m, 56.21m, 56.31m, 56.41m, 56.51m, 56.61m, 56.71m, 56.81m, 56.91m, 57.01m, 57.11m, 57.21m, 57.31m, 57.41m, 57.51m, 57.61m, 57.71m, 57.81m, 57.91m, 58.01m, 58.11m, 58.21m, 58.31m, 58.41m, 58.51m, 58.61m, 58.71m, 58.81m, 58.91m, 59.01m, 59.11m, 59.21m, 59.31m, 59.41m, 59.51m, 59.61m, 59.71m, 59.81m, 59.91m, 60.01m, 60.11m, 60.21m, 60.31m, 60.41m, 60.51m, 60.61m, 60.71m, 60.81m, 60.91m, 61.01m, 61.11m, 61.21m, 61.31m, 61.41m, 61.51m, 61.61m, 61.71m, 61.81m, 61.91m, 62.01m, 62.11m, 62.21m, 62.31m, 62.41m, 62.51m, 62.61m, 62.71m, 62.81m, 62.91m, 63.01m, 63.11m, 63.21m, 63.31m, 63.41m, 63.51m, 63.61m, 63.71m, 63.81m, 63.91m, 64.01m, 64.11m, 64.21m, 64.31m, 64.41m, 64.51m, 64.61m, 64.71m, 64.81m, 64.91m, 65.01m, 65.11m, 65.21m, 65.31m, 65.41m, 65.51m, 65.61m, 65.71m, 65.81m, 65.91m, 66.01m, 66.11m, 66.21m, 66.31m, 66.41m, 66.51m, 66.61m, 66.71m, 66.81m, 66.91m, 67.01m, 67.11m, 67.21m, 67.31m, 67.41m, 67.51m, 67.61m, 67.71m, 67.81m, 67.91m, 68.01m, 68.11m, 68.21m, 68.31m, 68.41m, 68.51m, 68.61m, 68.71m, 68.81m, 68.91m, 69.01m, 69.11m, 69.21m, 69.31m, 69.41m, 69.51m, 69.61m, 69.71m, 69.81m, 69.91m, 70.01m, 70.11m, 70.21m, 70.31m, 70.41m, 70.51m, 70.61m, 70.71m, 70.81m, 70.91m, 71.01m, 71.11m, 71.21m, 71.31m, 71.41m, 71.51m, 71.61m, 71.71m, 71.81m, 71.91m, 72.01m, 72.11m, 72.21m, 72.31m, 72.41m, 72.51m, 72.61m, 72.71m, 72.81m, 72.91m, 73.01m, 73.11m, 73.21m, 73.31m, 73.41m, 73.51m, 73.61m, 73.71m, 73.81m, 73.91m, 74.01m, 74.11m, 74.21m, 74.31m, 74.41m, 74.51m, 74.61m, 74.71m, 74.81m, 74.91m, 75.01m, 75.11m, 75.21m, 75.31m, 75.41m, 75.51m, 75.61m, 75.71m, 75.81m, 75.91m, 76.01m, 76.11m, 76.21m, 76.31m, 76.41m, 76.51m, 76.61m, 76.71m, 76.81m, 76.91m, 77.01m, 77.11m, 77.21m, 77.31m, 77.41m, 77.51m, 77.61m, 77.71m, 77.81m, 77.91m, 78.01m, 78.11m, 78.21m, 78.31m, 78.41m, 78.51m, 78.61m, 78.71m, 78.81m, 78.91m, 79.01m, 79.11m, 79.21m, 79.31m, 79.41m, 79.51m, 79.61m, 79.71m, 79.81m, 79.91m, 80.01m, 80.11m, 80.21m, 80.31m, 80.41m, 80.51m, 80.61m, 80.71m, 80.81m, 80.91m, 81.01m, 81.11m, 81.21m, 81.31m, 81.41m, 81.51m, 81.61m, 81.71m, 81.81m, 81.91m, 82.01m, 82.11m, 82.21m, 82.31m, 82.41m, 82.51m, 82.61m, 82.71m, 82.81m, 82.91m, 83.01m, 83.11m, 83.21m, 83.31m, 83.41m, 83.51m, 83.61m, 83.71m, 83.81m, 83.91m, 84.01m, 84.11m, 84.21m, 84.31m, 84.41m, 84.51m, 84.61m, 84.71m, 84.81m, 84.91m, 85.01m, 85.11m, 85.21m, 85.31m, 85.41m, 85.51m, 85.61m, 85.71m, 85.81m, 8